

Bank Debit & Credit Analysis

SQL Queries with Simple Explanations

Use these explanations to understand and present the SQL analysis in your project.

1. Total Credit

```
SELECT SUM(Amount) AS Total_Credit
FROM bank_transactions
WHERE Transaction_Type = 'Credit';
```

Explanation: Calculates the total amount of all credit transactions. SUM adds the Amount values, while WHERE keeps only transactions whose type is Credit.

2. Total Debit

```
SELECT SUM(Amount) AS Total_Debit
FROM bank_transactions
WHERE Transaction_Type = 'Debit';
```

Explanation: Calculates the total amount of all debit transactions by filtering the data to Debit transactions and adding their Amount values.

3. Net Transaction Amount

```
SELECT
    SUM(CASE WHEN Transaction_Type = 'Credit' THEN Amount ELSE 0 END)
    -
    SUM(CASE WHEN Transaction_Type = 'Debit' THEN Amount ELSE 0 END)
    AS Net_Transaction_Amount
FROM bank_transactions;
```

Explanation: Calculates the difference between total Credit and total Debit. CASE WHEN separates the two transaction types, and subtracting Debit from Credit gives the net transaction amount.

4. Debit-Credit Ratio

```
SELECT
    SUM(CASE WHEN Transaction_Type = 'Debit' THEN Amount ELSE 0 END)
    /
    NULLIF(SUM(CASE WHEN Transaction_Type = 'Credit' THEN Amount ELSE 0 END), 0)
    AS Debit_Credit_Ratio
FROM bank_transactions;
```

Explanation: Calculates how much debit exists for every unit of credit. NULLIF prevents a division-by-zero error if total credit is zero.

5. Transaction per Month

```
SELECT
    YEAR(Transaction_Date) AS Year,
    MONTH(Transaction_Date) AS Month,
    COUNT(*) AS Transaction_Count
FROM bank_transactions
GROUP BY YEAR(Transaction_Date), MONTH(Transaction_Date)
ORDER BY Year, Month;
```

Explanation: Counts transactions month by month. YEAR and MONTH extract the date parts, GROUP BY creates monthly groups, and ORDER BY displays them chronologically.

6. Transaction Amount by Branch

```
SELECT
    Branch,
    SUM(Amount) AS Total_Transaction_Amount
FROM bank_transactions
GROUP BY Branch
ORDER BY Total_Transaction_Amount DESC;
```

Explanation: Shows the total transaction amount for each branch. GROUP BY creates one result for each branch, and DESC puts the highest amount first.

7. High Transaction Flag

```
SELECT
    High_Transaction_Flag,
    COUNT(*) AS Transaction_Count,
    SUM(Amount) AS Total_Amount
FROM bank_transactions
GROUP BY High_Transaction_Flag;
```

Explanation: Summarizes transactions according to the High Transaction Flag. It shows both the number of transactions and their total amount for each flag category.

8. Risk Flag

```
SELECT
    Risk_Flag,
    COUNT(*) AS Transaction_Count,
    SUM(Amount) AS Total_Amount
FROM bank_transactions
GROUP BY Risk_Flag;
```

Explanation: Groups transactions by risk category, such as High Risk or Normal, and calculates the count and total amount for each category.

9. Transaction Volume by Bank

```
SELECT
    Bank_Name,
    COUNT(*) AS Transaction_Volume,
    SUM(Amount) AS Total_Transaction_Amount
FROM bank_transactions
GROUP BY Bank_Name
ORDER BY Transaction_Volume DESC;
```

Explanation: Compares banks by transaction volume and total transaction amount. COUNT measures the number of transactions, while SUM measures their monetary value.

10. Transaction Method

```
SELECT
    Transaction_Method,
    COUNT(*) AS Transaction_Count,
    SUM(Amount) AS Total_Amount
FROM bank_transactions
GROUP BY Transaction_Method
ORDER BY Total_Amount DESC;
```

Explanation: Analyzes transaction methods such as Bank Transfer, Credit Card, and Debit Card. It shows transaction count and total amount for each method.

11. Credit vs Debit by Month

```
SELECT
    YEAR(Transaction_Date) AS Year,
    MONTH(Transaction_Date) AS Month,
    SUM(CASE WHEN Transaction_Type = 'Credit' THEN Amount ELSE 0 END) AS Total_Credit,
    SUM(CASE WHEN Transaction_Type = 'Debit' THEN Amount ELSE 0 END) AS Total_Debit
FROM bank_transactions
GROUP BY YEAR(Transaction_Date), MONTH(Transaction_Date)
ORDER BY Year, Month;
```

Explanation: Compares Credit and Debit amounts for every month. CASE WHEN separates Credit and Debit inside each monthly group.

12. Bank-wise Credit and Debit

```

SELECT
    Bank_Name,
    SUM(CASE WHEN Transaction_Type = 'Credit' THEN Amount ELSE 0 END) AS Total_Credit,
    SUM(CASE WHEN Transaction_Type = 'Debit' THEN Amount ELSE 0 END) AS Total_Debit
FROM bank_transactions
GROUP BY Bank_Name
ORDER BY Bank_Name;

```

Explanation: Shows Credit and Debit amounts separately for every bank. This helps compare the transaction behavior of different banks.

13. Branch-wise Credit and Debit

```

SELECT
    Branch,
    SUM(CASE WHEN Transaction_Type = 'Credit' THEN Amount ELSE 0 END) AS Total_Credit,
    SUM(CASE WHEN Transaction_Type = 'Debit' THEN Amount ELSE 0 END) AS Total_Debit
FROM bank_transactions
GROUP BY Branch;

```

Explanation: Shows the Credit and Debit amounts for each branch, allowing branch-level comparison of money coming in versus money going out.

14. Suspicious Transactions

```

SELECT *
FROM bank_transactions
WHERE Suspicious_Transaction = 'Yes';

```

Explanation: Returns only transactions marked as suspicious. This can be used to identify transactions that may require additional monitoring or investigation.